SNS COLLEGE OF TECHNOLOGY

23ITP202 MINI PROJECT- I



ELDERGUARD AI: CONNECTED ELDER CARE PLATFORM

Problem Statement: Elder Safety, Health Monitoring and Connected Care

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PROBLEM STATEMENT · ELDER SAFETY

Keeping Elders Safe: A Connected-Care Approach

Many older adults live apart from their families. Missed medicines, unnoticed changes in health and slow emergency response put them at risk, while family, doctors, pharmacies and hospitals each work from separate information. We built one connected platform where everyone in an elder's care circle sees the same live record and can act in time.

WHAT HAPPENS TODAY

The 8:00 AM dose is missed. Blood pressure is not logged for days. A fall at night and no one knows.

WHAT ELDERGUARD DOES

A reminder rings until it is answered, and the caregiver sees the dose logged live. One tap on SOS alerts family first, then a second contact, the hospital desk and emergency services if nobody responds.

Understanding the Empathy Phase: Connecting with Elders and Caregivers

Deep Elder & Caregiver Research

Start with genuine curiosity about the lives of older adults and the family members who worry about them. Understand daily routines, fears of being alone and how comfortable they are with technology.

Building Emotional Connections

Go beyond demographics. What makes an elder hesitate to ask for help? What keeps a caregiver awake at night? What would make both feel in control?

- Interview elders and family caregivers
- Observe medicine routines at home
- Build empathy maps of feelings and needs

- Medicine-routine diary studies
- Contextual enquiry in homes and care centres
- Journey mapping of an emergency



Empathy map of a family caregiver. Illustrative persona; interviews with real elders and caregivers are planned.

Define the Problem: Identifying Core Pain Points in Elder Care

Missed Medicines

Several daily medicines are easy to forget, and the family finds out late, if at all.

Slow Emergency Response

A fall or sudden illness can go unnoticed. Help depends on someone happening to call.

Disconnected Information

Family, doctor, pharmacy and hospital each hold a piece, so prescriptions, vitals and history are not shared.

Hard-to-Use Technology

Small text, crowded screens and complex steps leave out the people who need help most.

The problem statement emerges from synthesis: "Elders and their families need a simple, connected way to take medicines on time, share health information and get help fast, so older adults can live independently with confidence."

Ideate Creative Solutions: Brainstorming Elder Care Features

Reminders That Get Answered

One gentle alarm per dose time. A tap logs the dose, and repeated misses alert the caregiver.

One-Tap SOS with Escalation

Family first, then a second contact, the hospital desk and emergency services on a timer.

Vitals and AI Companion

Track blood pressure, sugar, oxygen and heart rate, and chat with Mevi, a patient AI companion.

More ideas from the brainstorm

- Prescription scanner that turns a photo into a medicine schedule
- Doctor appointments with queue tokens
- Doctor-to-pharmacy e-prescriptions
- Five role-based portals with a large-text mode

Innovation thrives when we encourage wild ideas, defer judgement and build on each other's concepts. Generate quantity first, then refine for safety and quality.

Proposed Solution: How ElderGuard Works End to End

1. Capture

Elders log medicines and vitals; doctors write prescriptions

2. Connect

One shared record links elder, caregiver, doctor, pharmacy and hospital

3. Alert

Reminders ring, unusual readings are flagged, SOS goes out in one tap

4. Escalate

Caregiver now, second contact at 2 min, hospital at 5, emergency services at 10

5. Review

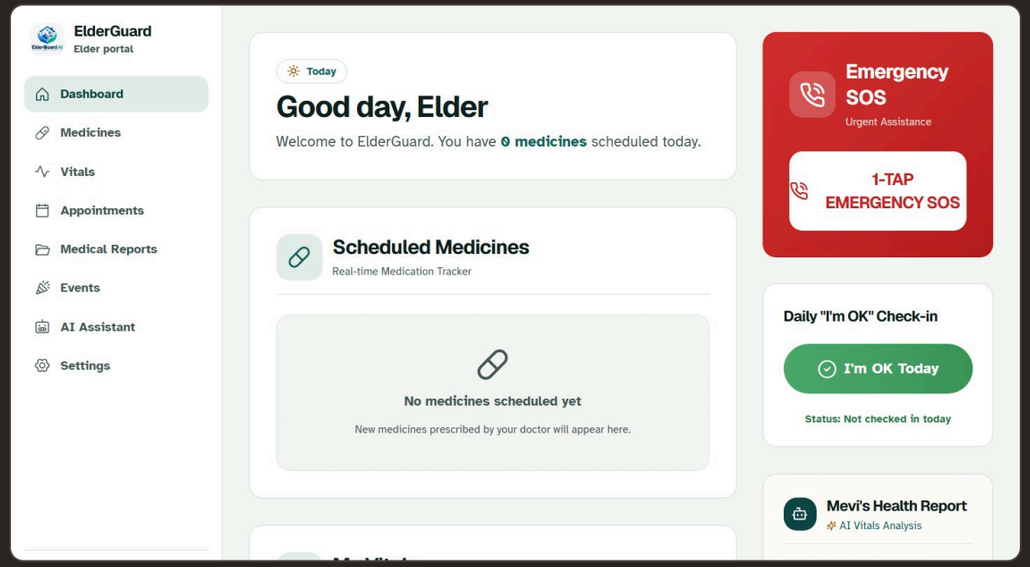
History and records let family and doctors act on what happened

Safety guardrails built in

- Role-based access: each user sees only the elders they are linked to
- ElderGuard alerts and informs; it does not diagnose or replace a doctor or emergency services

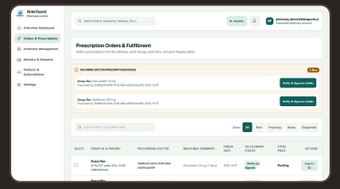
Prototype Development: Building Tangible Solutions for User Testing

ElderGuard AI · working prototype

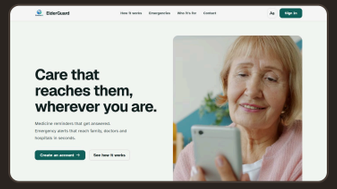




Caregiver portal



Pharmacy queue



Landing page

Design System

Warm paper background, one teal accent, large readable type and a text-size control for low vision.

Interactive Portals

Five role-based portals (elder, caregiver, doctor, hospital, pharmacy) running on real data.

Minimum Viable Product

A working React and FastAPI app: reminders, SOS escalation, vitals, and prescriptions that reach the pharmacy.

Prototypes are learning tools. Build fast, fail fast and iterate based on what real users show you.

Testing Phase: Verifying the Prototype and Planning User Trials

Automated Testing

An automated browser opens every portal page, the demo story is checked end to end, and each screen is scanned against WCAG 2.1 accessibility rules.

Key Testing Methods

End-to-end tests (Playwright): 22 tests across all five portals and phone size

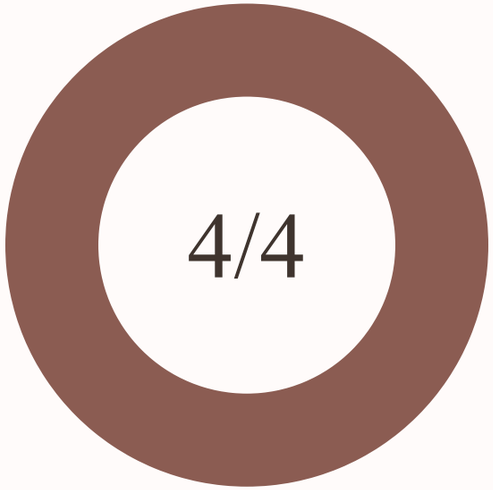
Accessibility audit (axe): 0 serious violations against WCAG 2.1 A/AA

SOS escalation drill: all 4 tiers fired in order in a timed run

Planned: teach-back and think-aloud sessions with real elders and caregivers



Automated tests passing



SOS tiers fired in the timed drill

Testing is discovery, not validation. Automated checks show that it works; sessions with real elders will show whether it helps.

Use Case: How Connected Care Protects an Elder at Home

Initial Challenge

Mrs. Anaya Rao, 77, lives alone and takes three medicines a day. Her daughter Maya lives in another city and worries about missed doses and emergencies.

1

2

Empathy Insights

Anaya does not want to be a burden and Maya cannot call every hour. They want reassurance that does not feel like surveillance.

Design Response

A reminder rings and Anaya taps Taken; Maya sees it live. Dr. Mehta prescribes in the app and the order reaches the pharmacy. One tap on SOS alerts Maya first, then the hospital.

3

4

Expected Impact

Fewer missed doses, faster response and less worry for the family. To be measured in a pilot.

Illustrative scenario based on this project's demo data; impact is a target to be validated.

Integration Challenges: Balancing Elder Needs with Technical and Clinical Constraints

Privacy & Compliance

Health data is highly sensitive. It needs consent, encryption and access logs in line with India's DPDP Act. Role-based access is in place; consent and audit features come next.

Emergency Reliability

SOS must never stall. Alerts are in-app and push today; SMS and voice-call providers and a durable job queue are needed for production.

Data & Integration

Records come from elders, doctors, pharmacies and hospitals. One shared data model and clean APIs keep them consistent.

Access & Performance

Older users often have older phones and slow networks. Large text and simple screens are prioritised; an installable app is next.

Safety first

ElderGuard alerts and informs; it does not diagnose or replace a doctor or emergency services.

Successful integration needs continuous dialogue between clinicians, designers and engineers. Constraints often spark creative solutions when teams collaborate early.

Success Metrics: Measuring the Impact of ElderGuard

22/22

Tests Passing

Automated end-to-end checks across all five portals

0

Accessibility Issues

Serious issues found by an axe audit (WCAG 2.1 A/AA) on every page

4

Escalation Tiers

Caregiver, second contact, hospital, emergency services, verified in a timed drill

90%+

Doses On Time

Pilot target for doses logged when due, to be validated

The first three are measured on the prototype; the last is a target for the pilot, to be validated with real users

Qualitative Indicators of Success

- Caregivers report less worry and fewer check-in calls
- Elders use the SOS button and large-text mode without help
- Doctors and pharmacies adopt the shared prescription flow
- Shorter time from SOS to first response

Future Enhancements: Expanding ElderGuard's Reach and Reliability

Phase 1

SMS & Voice-Call Alerts

Reach caregivers and hospitals even without the app, with delivery receipts and retries.

Phase 1

Verified Doctors & Hospitals

Licence checks before a clinical account can issue prescriptions.

Phase 2

Multilingual Voice Assistant

Speak and listen in Tamil, Hindi and other regional languages.

Phase 2

Installable Mobile App

Push notifications, offline mode and an always-ready SOS button.

Phase 3

Wearable Integration

Read heart rate and blood pressure, and detect falls, from smart watches.

Phase 3

EHR & ABDM Integration

Pull records securely from hospitals and ABHA health IDs, with consent.

Roadmap: Phase 1 reliable alerts and trusted accounts → Phase 2 voice and mobile → Phase 3 wearables and health-record integration.

Next Steps: Taking ElderGuard from Prototype to Pilot

Validate with Elders and Families

Run usability sessions with older adults and caregivers, and refine the screens from what they struggle with.

Harden for Real Use

Add SMS and call alerts, secure secrets and logins, privacy consent and monitoring.

Pilot and Scale

Partner with a clinic or care home for a small pilot, measure results, then expand.

In Summary

Connected information

→ faster response

→ safer elders

→ peace of mind

Start small, measure impact, and scale what works.

Elder care is an ongoing commitment to staying close to the people we serve and improving relentlessly.

Thank you!

ElderGuard AI: Care that reaches them, wherever you are